# Challenge 1: finding the real cause

## #!/usr/bin/env python3

The substring starts at offset 6. The old rule checked offset 1.

OLD RULE: WRONG POSITION

```
/bin/env python
at offset 1
```

**application/x-sh**

FIXED RULES: OFFSET 0

```
#!/usr/bin/env python
#! /usr/bin/env python
```

**text/x-python**

Two literal prefixes follow the existing direct-path Python rules.

# Challenge 2: proving the fix

## 01 Separate content from filename hints

Test each input without a filename, then with script.py.
This checks the content rule and the effect of metadata.

## 02 Keep working behavior intact

Check direct Python paths, Shell scripts and an input
with a newline before the shebang.

## 03 Same 18 focused checks: 7 failures before, 0 after

All 18 passed after the fix. These are local results.
They do not establish remote CI or whole-project validation.

# Negative tests catch false positives

## x/bin/env python

x is at offset 0. /bin/env python begins at offset 1.

OLD RULE: ACCIDENTAL MATCH

Matches the text after x
Mistakes ordinary text for Python

**text/x-python**

EARLIER FIX: FILE-START CHECK

No #! prefix at the start
The Python rule does not match

**text/plain**

Kyubin checked the earlier fix separately. A permanent regression test is deferred.

# What I learned and would change

| | |
|---|---|
| **Understand rule interactions** | Trace competing rules and write the expected behavior before changing the implementation. |
| **Plan broader coverage earlier** | Build an initial test matrix for Python, Shell, filename hints and boundary cases. |
| **Preserve useful review checks** | Turn probes such as x/bin/env python into permanent automated regression tests. |
| **Keep the useful workflow** | Continue reproducing first and comparing the same checks before and after a focused fix. |